Homework 07

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Question 1: LDA

The predictor matrix has dimension 462×9, the matrix of class means is 2×9, the pooled covariance matrix is 9×9, and the score matrix is 462×2. The empirical class priors are $\widehat{\pi}_0 = 0.6537$ and $\widehat{\pi}_1 = 0.3463$.

(a) Estimated components

Table 1: Estimated class means (famhist: Absent = 1, Present = 2).

class	sbp	tobacco	ldl	adiposity	famhist	typea	obesity	alcohol	age
0	135.4603	2.6347	4.3442	23.9691	1.3179	52.3675	25.7375	15.9314	38.8543
1	143.7375	5.5249	5.4879	28.1202	1.6000	54.4938	26.6229	19.1452	50.2938

Using $n - K = 460$ in the denominator, the pooled covariance estimate is shown in two column blocks below.

Table 2: Pooled covariance matrix, columns 1–5.

	sbp	tobacco	ldl	adiposity	famhist
sbp	405.4348	14.5852	4.5812	49.1644	0.3370
tobacco	14.5852	19.2426	0.7632	7.5381	0.0158
ldl	4.5812	0.7632	4.0006	6.0327	0.0919
adiposity	49.1644	7.5381	6.0327	56.7529	0.4328
famhist	0.3370	0.0158	0.0919	0.4328	0.2258
typea	-15.5877	-2.0573	0.3446	-5.3096	0.0811
obesity	18.9388	1.8335	2.6601	22.7079	0.1840
alcohol	64.4004	20.5170	-2.5329	16.1190	0.7685
age	95.1339	22.7654	6.4789	60.5087	0.9974

	typea	obesity	alcohol	age
sbp	-15.5877	18.9388	64.4004	95.1339
tobacco	-2.0573	1.8335	20.5170	22.7654
ldl	0.3446	2.6601	-2.5329	6.4789
adiposity	-5.3096	22.7079	16.1190	60.5087
famhist	0.0811	0.1840	0.7685	0.9974
typea	95.5656	2.6401	7.9600	-20.2783
obesity	2.6401	17.6154	4.6894	15.6970
alcohol	7.9600	4.6894	598.2766	27.8859
age	-20.2783	15.6970	27.8859	184.1321

(b) Manual scores and fitted labels

For each observation I evaluated the two stated discriminant scores directly and chose the larger one. The observed-by-predicted confusion matrix is

	Predicted	
Observed	0	1
0	258	44
1	73	87

Thus, 345 of 462 observations were classified correctly, for an in-sample misclassification rate of $117/462 = 0.2532$.

(c) Package LDA

`MASS::lda` with the same predictors and empirical priors produced the same confusion matrix. Therefore, the package and manual results match. Their stored label types differ, but the predicted class values are identical.

Question 2: QDA and marginal screening

I ranked pixels by their sample variances in the training set only and applied the same selected columns, without standardization, to the test set.

(a) Forty predictors

Table 3: QDA test confusion matrix using the 40 highest-variance pixels. Rows are observed and columns are predicted.

	0	1	2	3	4	5	6	7	8	9
0	319	0	24	6	2	2	1	0	5	0
1	0	238	5	0	11	0	7	0	3	0
2	8	0	168	8	8	3	0	0	3	0
3	5	0	4	128	0	14	0	0	15	0
4	0	2	13	1	150	1	0	1	7	25
5	12	0	3	5	1	131	0	0	6	2
6	0	0	8	0	1	8	148	0	5	0
7	1	0	6	1	6	0	0	124	2	7
8	1	0	10	12	3	2	0	0	137	1
9	0	0	1	2	7	1	1	4	9	152

The test misclassification rate is **0.1555** (312 errors among 2,007 test digits).

(b) Does it work?

Yes. QDA fits without a singular-covariance failure and classifies 84.45% of the test digits correctly. Screening to 40 predictors keeps the number of within-class covariance parameters manageable relative to each class sample size. The selected high-variance pixels also tend to occur where digit images differ, so they retain useful signal. The method is not perfect: marginal variance ignores the response and can retain variable pixels that are not the most discriminative, while discarding jointly useful pixels.

(c) Ten predictors

Table 4: QDA test confusion matrix using the 10 highest-variance pixels. Rows are observed and columns are predicted.

	0	1	2	3	4	5	6	7	8	9
0	298	0	11	1	1	16	28	4	0	0
1	0	241	10	0	3	1	1	0	2	6
2	19	2	95	14	14	3	22	21	5	3
3	5	0	6	103	3	42	0	0	7	0
4	1	5	14	0	95	2	4	15	17	47
5	15	0	6	25	4	96	4	2	8	0
6	41	0	15	4	0	18	78	7	5	2
7	4	1	6	0	6	0	1	122	0	7
8	1	2	11	36	12	11	4	1	74	14
9	0	2	9	2	20	0	0	16	10	118

The 10-predictor error is **0.3423** (687/2,007), 18.68 percentage points higher than the 40-predictor error. Although estimating each covariance matrix with 10 variables has lower variance, the aggressive screening removes too much information about digit shape. Here, that increase in bias outweighs the variance reduction.

Question 3: Compare LDA and QDA

I used seed **43207**. Each training set had 100 observations per class and each test set had 1,000 per class; LDA and QDA used the same data within every setting and equal class priors.

(a) One experiment per ρ

Test misclassification rates

ρ	LDA	QDA
0.0	0.1695	0.1930
0.2	0.1685	0.1820
0.4	0.1740	0.2075
0.6	0.1690	0.1945
0.8	0.1695	0.1375
0.9	0.1765	0.0990

(b) Fifty repetitions

Average test error

ρ	LDA	QDA
0.0	0.1694	0.1975
0.2	0.1706	0.1976
0.4	0.1715	0.1936
0.6	0.1704	0.1771
0.8	0.1673	0.1344
0.9	0.1645	0.0989

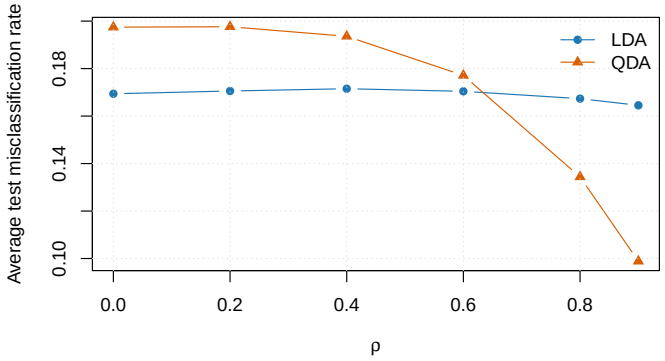


Figure 1: Average test error across 50 repetitions. Each point uses new training and test samples; both methods share samples within a repetition.

(c) Interpretation

At $\rho = 0$, the class covariance matrices are equal, so LDA's pooled covariance assumption is correct. LDA has lower error because it estimates fewer parameters. As ρ grows, the two classes have increasingly different covariances and the true boundary becomes nonlinear. LDA cannot represent that boundary, so its average error stays near 0.17; QDA pays an estimation-variance cost at small ρ but eventually benefits from its correct class-specific covariances, reaching 0.0989 at $\rho = 0.9$.

The eight noise variables contain no population information about class, but finite samples create accidental mean and covariance differences. Both methods estimate those irrelevant quantities; QDA estimates them separately by class and therefore incurs more variance. This explains part of QDA's disadvantage when the simpler LDA structure is adequate.

Question 4: LDA/QDA analysis skill

One-page SAheart analysis report

Methods. Using seed **43207**, I stratified the 462 observations into 347 training observations and an untouched test set of 115 (25%). On the training data I compared ordinary LDA, ordinary QDA, and PCA-LDA/PCA-QDA using 2–8 components. All candidates used the same five stratified folds and mean misclassification rate. Within every fold, centering, scaling, PCA loadings, priors, and classifier parameters were estimated using only the analysis portion. I selected the minimum-CV-error candidate, refit its preprocessing and classifier on all 347 training observations, and evaluated the test set once.

Table 5: Mean five-fold CV error. The selected minimum is bold.

Model	LDA	QDA
Ordinary (no PCA)	0.279	0.277
PCA, 2 PCs	0.320	0.303
PCA, 3 PCs	0.314	0.308
PCA, 4 PCs	0.300	0.297
PCA, 5 PCs	0.265	0.280
PCA, 6 PCs	0.268	0.279
PCA, 7 PCs	**0.262**	0.297
PCA, 8 PCs	0.271	0.277

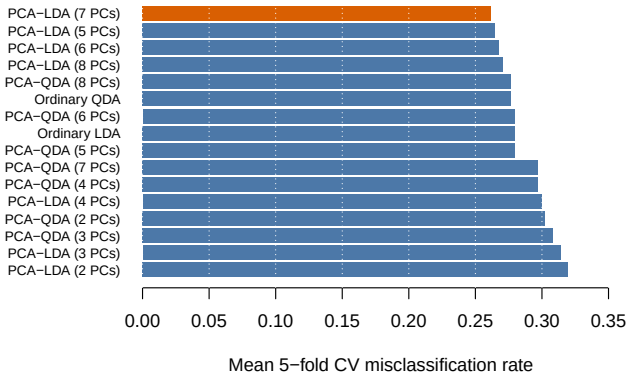


Figure 2: Shared-fold CV errors for all candidates; orange marks the selected model.

Final result and interpretation. Seven-component PCA-LDA was selected with CV error **0.262**. After refitting, its untouched-test error was **0.261** (30/115). The observed-by-predicted matrix was $\left[\begin{smallmatrix} 63 & 12 \\ 18 & 22 \end{smallmatrix}\right]$ for observed classes 0 and 1. Dimension reduction slightly improved CV error over ordinary LDA (0.279) and QDA (0.277), suggesting that removing low-variance directions reduced estimation noise while retaining most predictive structure. The single holdout is only one uncertain estimate, and both model families rely on Gaussian class structure; class-1 sensitivity was only $22/40 = 0.55$.

Skill definition and invocation

The final `SKILL.md` is **3,059 bytes**, below the 5,000-byte limit. I kept it outside the automatically loaded skill directory and set `allow_implicit_invocation: false` in its `agents/openai.yaml` configuration. I applied it with this explicit prompt:

> Use `$lda-qda-analysis` to analyze `data/saheart.csv`. Predict `chd` from all nine other variables, coding `famhist` as Absent = 1 and Present = 2. Seek the best predictive performance among the skill's allowed LDA/QDA models, include ordinary LDA and QDA, use seed 43207, and produce the required one-page report.

The complete contents of `SKILL.md` follow.

```
---
name: lda-qda-analysis
description: Compare LDA/QDA classifiers and produce a concise, reproducible prediction report. Use only when explicitly invoked with a
      dataset, response, and predictor specification.
---

# LDA/QDA Analysis

Analyze the supplied classification data using only linear discriminant analysis (LDA), quadratic discriminant analysis (QDA), or
      combinations of their predicted class probabilities. Seek strong out-of-sample predictive performance while keeping all model
      selection separate from final evaluation.

## Analysis requirements

- Confirm the response classes, requested predictors, sample size, missingness, and any user-specified coding. Do not silently add
      variables or change the target.
- Set and report one reproducible random seed. Before model comparison, reserve a stratified test set of about 20--25%; do not inspect its
      outcomes again until one model is selected.
- On the remaining data, compare ordinary LDA and ordinary QDA. When sample size and rank allow, also compare reasonable PCA-LDA and PCA-QDA
       candidates; a probability blend is optional. Keep every classifier based on LDA, QDA, or their combination.
- Use the same stratified cross-validation folds for every candidate. Choose settings only by mean validation misclassification rate, with
      a simple-model tie break.
- Prevent leakage: within each fold, estimate imputation, scaling, PCA/loadings, covariance matrices, class priors, and any blend weight
      using only that fold's analysis observations. Apply those fitted transformations unchanged to its validation observations.
- If a covariance estimate is singular or numerically unstable, report the failure instead of hiding it. A lower-dimensional PCA candidate
      may address instability.
- After selection, refit the chosen preprocessing and classifier on all non-test observations. Evaluate it once on the untouched test set.
      Report the test misclassification rate and confusion matrix; label rows as observed and columns as predicted.
- Clearly distinguish cross-validation results (model selection evidence) from the final test result (prediction-error estimate). Do not
      select or revise models from test performance.

## Deliverable

Produce a clearly readable analysis report that fits on at most one printed page. It must include:

1. a short methods paragraph naming the candidates, split, folds, metric, seed, and leakage controls;
2. one compact table of numerical cross-validation results, visibly marking the selected model;
3. one figure that compares candidate validation errors or otherwise directly supports selection;
4. the selected model's final test misclassification rate and observed-by-predicted confusion matrix; and
5. a brief interpretation of the main finding and limitations, including uncertainty from a single split and any assumption or stability
      concern.

Use concise captions, readable labels, and consistent rounding. Keep code outside the one-page report unless the user requests it, but
      retain reproducible analysis code as a separate working artifact when possible.
```